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Whole slide imaging of circulating tumor cells captured on a capillary microchannel device

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Liquid biopsy with circulating tumor cells (CTCs) can aid in cancer detection at early stages and determine whether a certain treatment is effective or not. However, existing CTC techniques focus on one or two aspects of CTC management including sampling, enrichment, enumeration, and treatment selection. This paper reports an integrated capillary microchannel device that allows efficient capturing of CTCs with a wide microchannel, rapid enumeration of captured CTCs with whole slide cell imaging, and *in situ* drug testing with captured CTCs. Blood is drawn into the microchannel whose height is appropriate to the diameter of cancer cells, while its width is a thousand times larger than the diameter of cancer cells. The inner bottom surface of the microchannel is modified with long chain polymers that have cell adhesive ends to efficiently capture CTCs from blood. With this design, cells including CTCs are forced to move through the polymer coated microchannel, and the chance of cell adhesive ends interacting with specific antigens overexpressed on surfaces of cancer cells is significantly enhanced without a channel blockage issue. Captured CTCs are enumerated with a whole slide imaging platform *via* dual LED autofocusing technology then exposed to anti-cancer drugs, followed by live/dead assay and fluorescence imaging. Given its straightforward, easy and powerless operation, this device with whole slide imaging will be useful for cancer diagnosis, prognosis and point-of-care treatments.

Introduction

Each year, 8 million people die of cancer, and 90% of cancer deaths are caused by metastatic cancers.[1,2] Key components of cancer metastasis are circulating tumor cells (CTCs), which are cancer cells that are released from the original tumor and enter blood circulation.[3,4] In so-named liquid biopsy, CTCs are considered as biomarkers for cancer diagnosis and prognosis, and a simple minimally invasive blood test can detect CTC levels in patients to diagnose cancers at early stages or to monitor treatment effects if cancers have been confirmed.[5–9]

A major challenge of CTC detection is that CTCs in blood of cancer patients are rare (10–1000 CTCs in 1 mL of blood) in a huge background of red blood cells.[10,11] Given the low concentration of CTCs in blood (especially at early stages), a variety of techniques have been used to pre-treat blood samples and enrich CTCs before detection.[12–14] CTC enrichment methods are based on differences in physical and biological properties between CTCs and healthy cells which include size, deformability, dielectric properties, and receptors that are overexpressed on surfaces of cancer cells. Among existing techniques for CTC enrichment, such as using magnetic micro- or nanoparticles,[15,16] microfiltration,[17] dielectric separation,[18] and flow cytometry,[19] microfluidic chips have attracted attention due to their low sample volume and possibility of process automation.[20–24] However, normal microfluidic devices with a narrow width tend to be blocked if blood samples are used directly, and thus blood samples have to be processed extensively to remove red blood cells before CTC enrichment. A CTC chip with thousands of vertical microcolumns can partially solve the blockage issue,[25] but CTCs captured through antigen–antibody interaction could be located at any height of each microcolumn (height of 500 μm), and the focus (and the field of view) of an optical microscope has to be constantly adjusted to find captured CTCs. Meanwhile, to increase capture efficiency, microfluidic chips should be modified with specific capture species such as anti-epithelial cell adhesion molecules, but the molecules are directly immobilized on substrates and are not flexible enough to interact with receptors overexpressed on surfaces of moving cancer cells. The devices, though designed to be low cost and field deployable, require expensive processes to make, and sophisticated accessories such as pumps, tubing and electricity to work, and therefore are accessible only to trained personnel at centralized facilities.

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Once captured, CTCs are detected with a variety of techniques such as fluorescence measurements,[26] photoacoustic methods,[27] Raman spectroscopy,[28] nanotechnology, and impedance measurements.[29] These detection techniques need either expensive labels or sophisticated instruments not available at point-of-use. Most importantly, even when the level of CTCs has been determined, patients still do not receive benefits from the enrichment and detection until a drug is prescribed or a change in dosage is implemented. However, there is a discontinuity among the major aspects of CTC management that prevents cancer patients from being treated in a timely manner. It may take up to a few days from blood sampling, CTC enrichment, enumeration or detection to treatment selection. During this period, CTCs in patients may have already rooted in distant locations, making subsequent treatment difficult or impossible. Ideally, the whole process of CTC management should be performed rapidly at point-of-care so that treatments can be started instantaneously once the changes in CTC levels in patients are confirmed.

To analyze captured CTCs, a large field of view and high-resolution imaging approach is needed to capture microscopic images of the entire chip. With the recent progress in Fourier ptychography,[30,31] it is possible to use a large field of view, low numerical aperture (NA) objective lens to acquire a high resolution image without mechanical scanning. This approach, however, is still in its early stage for high-resolution fluorescence imaging.[32,33] A straightforward solution for acquiring wide-field high-resolution image of the entire CTC chip is to perform whole slide imaging (WSI),[34] which has been adopted in digital pathology for better and faster prediction and prognosis of cancers and other diseases. It can be used for bright field, phase-contrast, and fluorescence imaging.[35–37] The WSI technique has not been demonstrated for whole slide CTC imaging. For the proposed CTC chip, one technical challenge for WSI is the autofocusing process, as CTCs are transparent under bright field illumination. A recently developed two-LED approach can be used to perform autofocusing of the transparent sample without fluorescence labeling,[37] so that a whole-slide phase-contrast image can be generated *via* oblique LED illumination.

Project rationale

This paper describes an integrated capillary microchannel device that allows direct blood sampling without sample pre-treatment, efficient capturing of CTCs with a wide microchannel, rapid enumeration of captured CTCs with whole slide cell imaging, and *in situ* drug testing with captured CTCs. Blood is drawn into the microchannel, whose height is appropriate to the diameter of cancer cells, while its width is a thousand times larger than that of cancer cells. The inner surfaces of the microchannel are modified with long chain polymers that have cell adhesive ends to efficiently capture CTCs from blood. Fig. 1 shows the scheme of an integrated microchannel device. All the cells, including CTCs, are forced to roll through the polymer-brushed microchannel, and the chance of cell adhesive ends interacting with specific antigens overexpressed on the surfaces of cancer cells is significantly enhanced, while there is no channel blockage issue. The captured CTCs are counted by using a wide field high resolution optical microscope. The CTCs are then treated with anti-cancer drugs, followed by live/dead assay and fluorescence imaging.

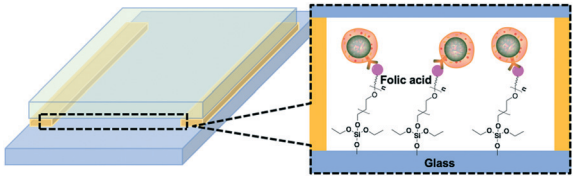

Fig. 1 Capillary microchannel device for efficient capture of CTCs with a cell monolayer, and cells including CTCs passing through the polymer brush modified microchannel.

This integrated microdevice has several distinctive advantages: (1) blood is extracted directly from patients without pre-treatment and drawn into the microchannel without sample handling and storage that may cause contamination and CTC loss. The sampling process is minimally invasive and can possibly be performed by patients themselves after training. (2) All the cells in blood are physically constrained to form a monolayer and roll through the microchannel device, which significantly decreases the kinetic barrier for cell transport from a bulk solution onto a capture substrate. (3) Cancer cells are carried by capillary flow in the microchannel to pass through a polymer brush that has cell adhesive terminals (folic acid). Folic acid terminals bind high affinity folate receptors overexpressed on tumor cell membranes, and isolate cancer cells among normal blood cells. The elastic nature of the polymer chain enables conformal contact and intimate interaction of terminal folic acid molecules with surface folate receptors on moving cells, which can enhance its capture selectivity to cancer cells. (4) The issue of channel blockage frequently seen in standard microfluidic devices is absent in the microchannel device with a millimeter to centimeter width. (5) The captured CTCs are enumerated using an imaging microchannel with a wide field high resolution optical microscope, and the response of the captured CTCs to anti-cancer drugs is determined *in situ* with live/dead assay and a fluorescence microscope. (6) The integrated device is made of low-cost materials, does not need expensive instruments and electricity to pump fluids, and is suitable for point-of-care applications.

Materials and methods

Chemicals and materials

Most chemicals, such as 3-triethoxysilylpropylamine, 12-mercaptododecanoic acid, 1-ethyl-3-3-dimethylaminopropyl-*N*′-ethylcarbodiimide, tris-carboxyethyl-phosphine hydrochloride,

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glutaric dialdehyde, ethanol, acetone, and dimethyl sulfoxide, are obtained from Aldrich. Poly(allylamine hydrochloride) with a molecular weight of 120–200 kDa and *N*-hydroxysuccinimide are obtained from Alfa Aesar. [Hydroxy(polyethyleneoxy)propyl]triethoxysilane (PEG-silane) is obtained from Gelest. Maleimide polyethylene glycol succinimidyl carboxymethyl ester (molecular weight of 3.5 kDa) is obtained from Creative PEGWorks. *N*,*N*-Dimethylformamide, phosphate buffered saline and glass slides are from VWR. Folic acid is from Enzo Life Sciences. A negative photoresist (SU-8) and its developer are obtained from MicroChem. Human blood is from Innovative Research. T lymphoblastoid cell line (CEM) is obtained from ATCC (American Type Cell Culture). An anti-cancer drug, cisplatin, is obtained from Sigma-Aldrich. Calcein AM, Hoechst 33342, RPMI 1640 medium supplemented with 10% fetal bovine serum, 100 units per mL penicillin, and 100 μg mL^{-1} streptomycin are from Thermo Fisher.

Glass surface modification

The bottom slide and top slide are modified separately. In order to modify the bottom slide of the microchannel (Fig. 2), a glass slide is cleaned by sonication in ethanol for 10 min followed by air drying. The glass slide is immersed in a 1 mM ethanol solution of 3-triethoxysilylpropylamine (APTES) to form an amine modified surface. Then, the glass slide is grafted with a layer of poly(allylamine hydrochloride) (PAH) to increase the density of amino groups by crosslinking the amine on APTES with glutaric dialdehyde (GA). The PAH coated slide is firstly immersed in 20 mM 12-mercaptododecanoic acid (MDA) in anhydrous dimethylformamide (DMF) that contains 1-ethyl-3-(3-dimethylaminopropyl)-*N*′-ethylcarbodiimide and hydroxysuccinimide (EDC/NHS) for 6 h, and then in 10 mg mL^{-1}maleimide polyethylene glycol succinimidyl carboxymethyl ester (Mal-PEG-NH_2) in phosphate buffered saline (PBS) that contains 40 mM tris(2-carboxyethyl)-phosphine hydrochloride (TCEP) at pH 7–7.5 for 1 h. Folic acid is dissolved in dimethyl sulfoxide (DMSO) at a concentration of 23 mg mL^{-1}, followed by activating carboxyl groups by adding EDC and NHS at a mole ratio of 2 : 1. After 24 h, the PEG modified glass is immersed in activated folic acid solution for 18 h to complete the reaction. The top glass slide is treated with oxygen plasma and immersed into a 5% ethanol solution of [hydroxy(polyethyleneoxy)propyl]-triethoxysilane (PEG-silane) overnight. The contact angle of the glass slide before and after each modification is measured with a Phoenix 150 contact angle measurement system to confirm the success of each step.

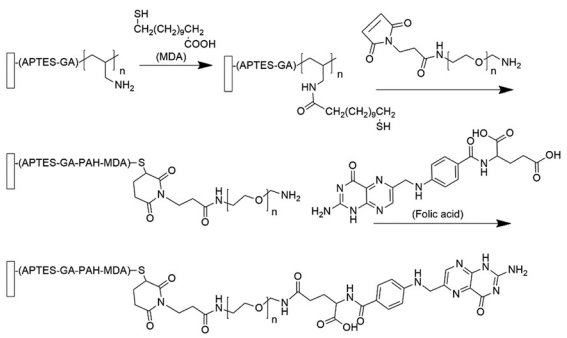

Fig. 2 Surface modification of the bottom glass slide.

Microchannel device fabrication

The top modified slide is coated with a thin film of SU8-2035. The samples are soft baked at 95 °C for 1 min on a hotplate, and then exposed to 365 nm UV light through a mask using a UV-KUB exposure system at a power density of 40 mW cm^{-2} for 10 s. The samples are then post-baked at 120 °C for 1 min and immersed in the developer to remove unexposed regions, followed by rinsing with water and blow-drying with air. In order to bond the top glass slide with the SU8 coated glass slide, pressure is applied using a press with bolted joints and a torque wrench. The clamped samples are kept under pressure and heated in an oven at 180 °C for 2 h.

Cancer cell capture and drug testing

CEM cells are cultured in RPMI 1640 medium supplemented with 10% fetal bovine serum (FBS), 100 units per mL penicillin, and 100 μg mL^{-1} streptomycin at 37 °C and 5% CO_2. Cell suspension or blood spiked with cancer cells is added at one side of the device and capillary force draws the fluid to completely fill the microchannel. After a certain time, a tissue paper is placed at the other end of the microchannel to remove the fluid and unbound cells. The filling process is repeated to fill the device with all of the sample. The microchannel is then washed with PBS by placing a tissue paper at the other end to generate continuous flow. The viability of the captured cells is tested by staining them with Calcein AM and Hoechst 33342. In order to test the efficacy of the chemical drugs on the captured cells, cisplatin is dissolved in PBS to reach a concentration of 10 μM and the solution is added at one end of the microchannel and incubated with the cells for a certain time. The conditions of the drug treated cells are examined with Calcein AM and propidium iodide (PI) staining, where live cells are in green and dead cells are in red. The cells captured in the microchannel device are imaged with a wide-field high resolution optical microscope (cell counting), an upright Olympus optical microscope (cell imaging), and an Olympus BX51 fluorescence microscope (for live/dead assay).

Results

Optical image of microchannel device

Fig. 3A shows the optical image of the capillary microchannel filled with an aqueous solution of rhodamine dye to highlight the capture region (dimensions of 5 cm × 3.6 cm). Fig. 3B shows the cross-section view of the microchannel end

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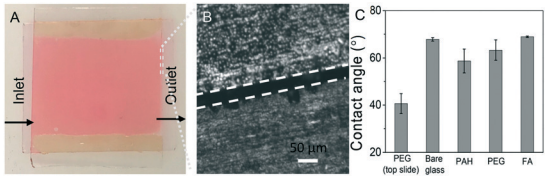

Fig. 3 Optical image of the microchannel device filled with rhodamine dye, which highlights the channel structure (A), cross-section view of the microchannel from one end (B), and contact angle of surface modified glass slides (C).

(highlighted by white dashed lines) with a height of 30 μm, where the microchannel height varies less than 8% along a 1 mm long distance. The uniform height of the microchannel ensures identical capture conditions for cells. The height of the microchannel has been adjusted by spinning the photoresist at different speeds. It is thus possible to make microchannels of different heights that are suitable for different types of cancer cells.

Cell capture on the modified glass substrate

The surface of the glass slide has been modified to have a polymer brush with cell-capturing ends. Fig. 3C shows the contact angles of the glass slides after each modification. Compared with the bare glass with a contact angle of 67.8°, the contact angle of the top slide modified with a monofunctional PEG (5 nm in length) is 40.6°. The contact angles of the slide are 58.7, 63.3, and 68.9° after sequential modification with PAH, long amine functionalized PEG (~30 nm in length), and folic acid, respectively. The slight increase in contact angle confirms the success of each modification and indicates the less hydrophobic nature of the capture slide. Fig. 4A–E show the optical micrographs of CEM cells attached on the glass slides after each modification, where loosely attached cells have been washed away and only those cells that remain on the surface after washing with PBS buffer with electrostatic attraction and specific antigen–antibody interaction are imaged. The bare glass slide can attract some cells at a surface density of 240 cells per mm^2 due to their negative charge in PBS (A). The top glass with PEG modification does not capture any cells due to the anti-adhesive nature of PEG (B). After PAH modification, the bottom glass slide is positively charged, and attracts more cells with a surface density of 2750 cells per mm^2 through electrostatic attraction with negatively charged lipids on the cell membrane (C). This non-specific electrostatic attraction is required to be blocked in order to differentiate tumor cells from blood cells. The PEG modification on the PAH modified slide significantly reduces the number of cells captured on the glass slide with a surface cell density of 16 cells per mm^2 (D), which confirms the effective blockage of electrostatic attraction. The glass slide that is modified with folic acid captures more cells with a surface cell density of 2600 cells per mm^2 (E).

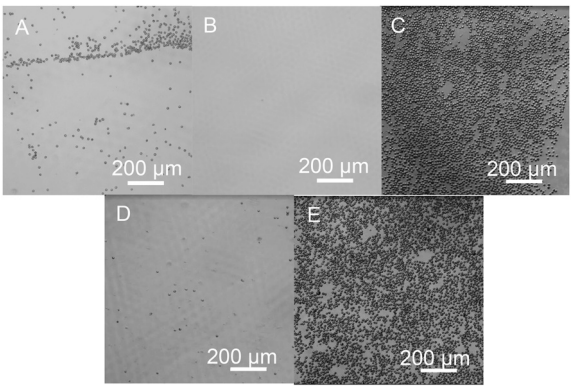

Fig. 4 CEM cells attached on the unmodified glass (A), PEG modified top glass (B), PAH modified glass (C), long PEG modified glass (D), and folic acid modified glass (E).

Cell capture with the surface modified microchannel device

The efficiency of the folic acid modified microchannel device in capturing CEM cells in PBS solution is tested by placing a droplet (volume of 50 μL) of cell suspension with a known cell concentration at the entry of the device, then washing away the suspension with PBS. Fig. 5A shows a fluorescence image of the captured CEM cells after staining with Calcein AM and Hoechst 33342, where the live cells are stained green and the dead ones are stained blue. The morphologies of the cells can be seen clearly, indicating that the cells maintain their integrity and are not disrupted by shear force when flowing through the device. The capture efficiency of the microchannel with different heights is derived from the average number of attached cells over 5 random locations and the number of all injected cells (Fig. 5B). As the height of the microchannel increases from 30 to 170 μm, the capture efficiency decreases from nearly 95% to 65%, confirming the effect of the microchannel height on cell capture. In an extreme case, a 100 μL droplet of cell suspension is placed on the top of the folic acid modified slide, and the capture efficiency is determined to be 1.3%. The low efficiency is due to the long settling time of cells, meaning that an advantage of the microchannel device for

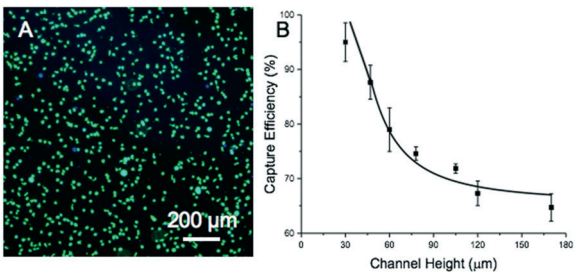

Fig. 5 Fluorescence image of the captured CEM cells after staining with Calcein AM and Hoechst 33342, where the live cells are stained green and the dead ones are stained blue (A). Cell capture efficiency of microchannels with different heights (B).

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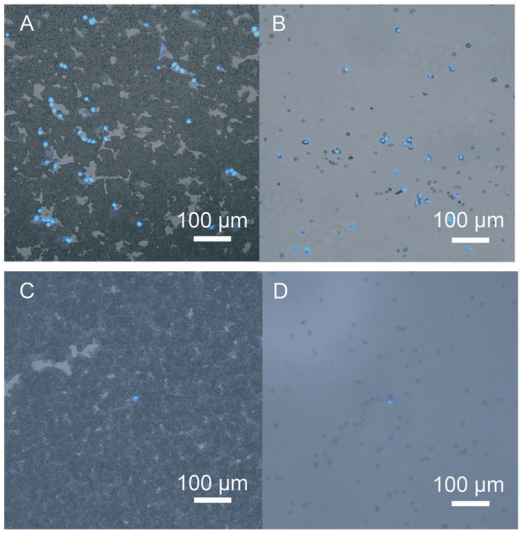

Fig. 6 Fluorescence images of the microchannel filled with blood, where the spiked CEM cells are fluorescently labeled before (A and C) and after (B and D) washing blood with the buffer solution. The initial concentrations of the CEM cells are 500 000 cells per mL (A) and 5000 cells per mL (C), respectively.

cell capture is that the cell settling time is effectively reduced.

Capture of blood containing CTCs with the surface modified microchannel device

In order to test the usage of the microchannel in capturing CTCs in blood, the CEM cells are spiked into the human blood. Fig. 6C and D show the fluorescence images of the CEM cells captured on the folic acid modified microchannel device (30 μm height) at a concentration of 5×10^3 cells per mL. Once a CEM cell is captured, the site is occupied and red blood cells surround the captured cell (Fig. 6C). The wide microchannel device prevents the CEM cells and red blood cells from blocking the channel. Fig. 6B and D show the images collected after red blood cells are washed away with PBS, and more than 99% red blood cells have been washed away. The capture efficiency of the device has been calculated to be 88.3 ± 14.6% for the spiked CEM cells. The reduced capture efficiency in blood compared to that in the buffer solution is likely due to the increased viscosity of blood and increased collision of the CEM cells and red blood cells.

Whole slide imaging of captured CTCs

A WSI system is used to acquire the image of the entire CTC chip. The system is built based on a motorized Olympus upright microscope with an NA = 0.75 objective lens. The autofocusing process is based on the recently developed dual-LED approach.[37] In the implementation, LED elements are placed at the back focal plane of the microscope condenser lens and they illuminated the sample with oblique incident angles. These LED elements can be treated as partially coherent light sources and they can generate a coherent contrast of the transparent samples at out-of-focus positions without labeling. After the autofocusing process, the same LED elements are used for oblique illumination and generate a phase contrast of the transparent CEM cells. Fig. 7 shows the captured whole-slide phase-contrast image of the CEM cells and the acquisition time is about ~3 min without optimizing the scanning process. The number of the CEM cells in the drop of blood that is directly counted from the stitched images is approximately 580 cells per mL. Compared to traditional image stitching, WSI does not need mechanical scanning in two dimensions, providing higher workflow efficiency and finer features of cells. The cell nucleus and protrusions on the cell surface can be imaged, as shown in the zoomed in views in Fig. 7.

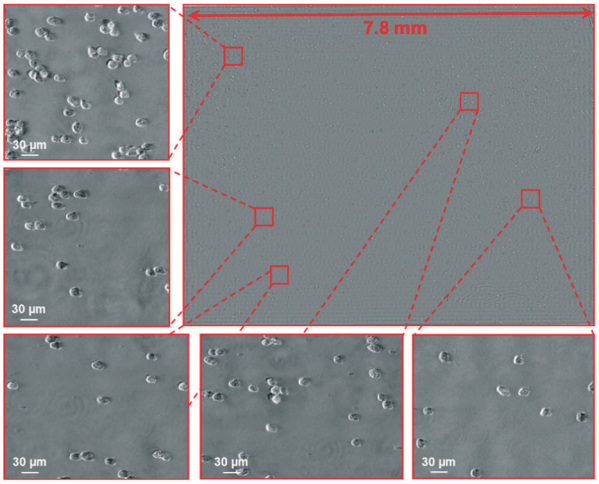

Fig. 7 The whole slide image of the entire CTC chip.

In situ drug testing with captured CTCs

The CEM cells captured within the microchannel have been directly exposed to an anti-cancer drug, *i.e.*, cisplatin, to evaluate the efficacy of the drug. Fig. 8A and B show the fluorescence images of the cells before and after exposure to cisplatin for over 24 h, followed by staining with Calcein AM and propidium iodide (PI). In the control experiment without the drug (Fig. 8A), more cells are stained green, meaning more cells are alive. In the drug treated samples, only few live cells can be observed in green and the majority of the cells

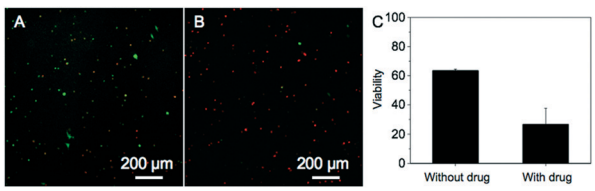

Fig. 8 Fluorescent images of the cells captured in the microchannel without drug treatment (A) and with drug treatment (B); the viabilities of cells without and with drug treatment, where the live cells are in green (Calcein AM staining) and the dead ones are in red (propidium iodide staining) (C).

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are stained red, indicating that the cells are dead after exposure to the drug for 24 h.

CTC capture with the capillary microchannel

Capillary force is responsible for pulling liquids into a microchannel, and determining how fast liquids fill a microchannel.[38] The liquid flow in the microchannel is characterized by low Reynolds numbers and can be treated as a laminar flow parallel to the surface. The time it takes for the liquid to fill up the channel is defined as filling time (t_f), which is related to the channel height (h) and calculatedas:[39]

$$t_f = \frac{3\,\mu L^2}{h\sigma\cos\theta} \tag{1}$$

where σ, θ, μ and L are the surface tension of the liquid, contact angle of the liquid on the substrate, viscosity of the liquid, and the channel length, respectively. The filling time is inversely proportional to the channel height (h). Therefore, higher channels reduce the filling time, and cells in higher channels take a longer time to settle. Assuming that all cells are injected at the entry of the microchannel and the time it takes for a cell to fall at the bottom is determined by their initial height in the microchannel, the velocity of cells during falling can be derived from the force balance between gravity, buoyant force and Stokes drag and calculated as:

$$F_G - F_B - F_S = m\frac{\partial v}{\partial t} \tag{2}$$

where $F_G = \frac{1}{6}\pi d^3\rho g$, $F_B = \frac{1}{6}\pi d^3\rho_0 g$, $F_S = 3\pi\eta v d$, and d, ρ, ρ_0, and g represent the cell diameter, cell density, liquid density and gravitational acceleration, respectively. The velocity of cells can then be derived by solving eqn (2) as shown below:

$$v = \frac{(\rho-\rho_0)gd^2}{18\eta}\left(1 - e^{-\frac{18\eta}{d^2\rho}t}\right) \tag{3}$$

Therefore, the distance traveled by one cell in the vertical direction can be derived from eqn (3):

$$S = \frac{(\rho-\rho_0)gd^2}{18\eta}t + \frac{\rho(\rho-\rho_0)gd^4}{324\eta^2}e^{-\frac{18\eta}{d^2\rho}t} - \frac{\rho(\rho-\rho_0)gd^4}{324\eta^2} \tag{4}$$

The terminal velocity v_0 is $\frac{(\rho-\rho_0)gd^2}{18\eta}$ when $F_1 - F_2 - F_3 = 0$.

Fig. 9 shows the time dependent sedimentation velocity and distance profile derived from MATLAB, where cells accelerate in a short time (10^{-4} s) until reaching terminal velocity and keep falling at a constant velocity, thus the falling velocity is considered constant. According to eqn (1), the filling time is calculated to be 6.69 s in a channel with a height of 30 μm and length of 50 mm when the viscosity and surface tension of the liquid (PBS) are 1.05 cp and 72 dynes per cm, respectively. The falling distance during liquid filling is then calculated to be 85 μm using:

$$h' = v_0 t_f = \frac{3\,\mu L^2 v_0}{h\sigma\cos\theta}.$$

This distance corresponds to the critical channel height. For $h' > 30$ μm, all cells will settle to the bottom during liquid filling, which is confirmed by the result of 95% capture efficiency during the filling process. However, if the channel height is larger than h', only some portions of cells that is below h' could be captured on the substrate during the filling process.

Discussion

Depending on the cancer stage, the concentration of CTCs in the blood of cancer patients is in the range of 10–1000 cells per mL, and the number of CTCs in a small droplet of blood (~0.1 mL) is in the range of 1–100 cells. Given the height of the microchannel of 30 μm and the two dimensional area of the microchannel of 50 mm × 67 mm, the volume is 0.1 mL. In principle, one droplet of blood from a fingertip can be analyzed to allow the early detection and prognosis of cancer based on CTCs. In the case of clinical samples, a larger volume of sample (~2 mL) should be used to provide reliable detection, and the blood samples can be filled in the microchannel by repetitive filling with the blood sample to cause a continuous flow. To reduce blood coagulation, an anti-coagulation agent may be added into the blood sample. Furthermore, a device of 5 cm wide can be divided into multiple parallel (5–10) channels of a few mm wide, so that multiple antibodies can be used to capture CTCs based on different surface receptors, which will enhance detection sensitivity and help to differentiate tumor sub-types.

The whole slide imaging method provides high resolution images of individual cells over a large area. There are other ways to image or count cells on wide fields. A mobile phone camera has been used to achieve whole slide cell imaging with a lens free imaging process algorithm, but the field of view is not large enough to image cells over an inch square slide. A laser cell scanner has been developed by Hewlett-Packard (HP) to count cells attached on a large plate, but the

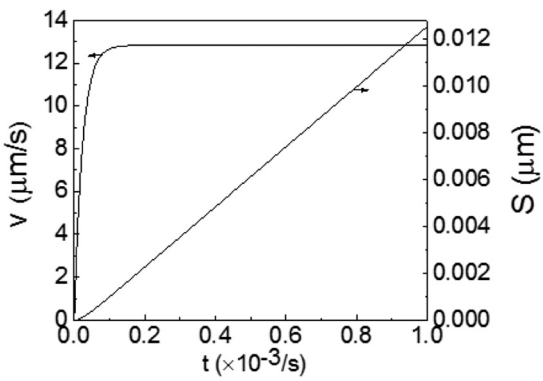

Fig. 9 Falling velocity and distance profile of cells during liquid filling of the microchannel.

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device is costly and can only give the number of cells. Cell imaging or counting can be achieved by fluorescently labelling the CTCs in the microchannels, where a microscopy video under ultraviolet illumination can be taken while the liquid is passing through the microchannel. A phone camera with a simple magnification attachment may be used to reduce the cost of detection and make it available to the general public.

Conclusions

Whole slide imaging is applied to a capillary microchannel device to assess captured circulating tumor cells (CTCs). CTCs spiked in human blood are captured efficiently within the capillary microchannel device that is modified with a polymer terminated with CTC capturing species. The blood is drawn in the microchannel by capillary force when the height of the microchannel is 30 μm. Once inside the microchannel, cells (including CTCs) are forced to roll through the polymer-brushed microchannel, and the chance of cell adhesive ends interacting with specific antigens overexpressed on the surfaces of cancer cells is dramatically enhanced. The capture efficiency increases with decreasing height of the channel due to rolling and tethering of cells in the microchannel. The captured CTCs are counted with a wide field high resolution microscope then treated with anti-cancer drugs, followed by live-dead assay and fluorescence observation. This microfluidic device is easy to make, low-cost, does not have a blocking issue, and allows the rapid capture and accurate detection of circulating tumor cells from a droplet of blood.

Conflicts of interest

There are no conflicts to declare.

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Notes and references

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